Top production cross sections in l-h

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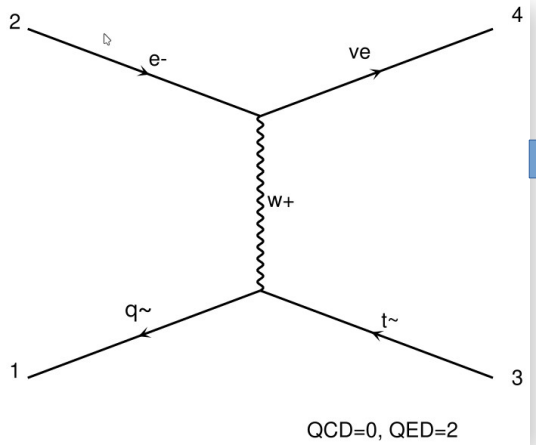
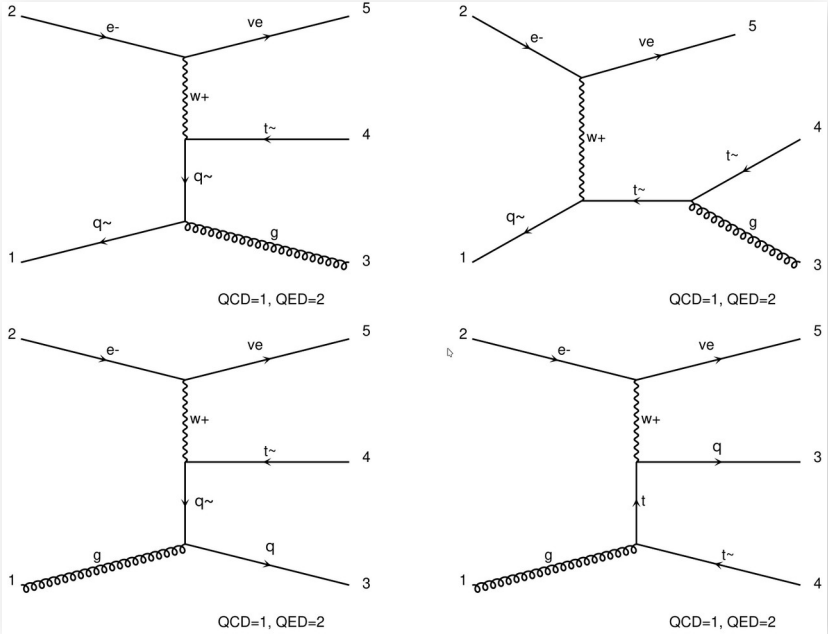
Single-top and top pair production at LHeC

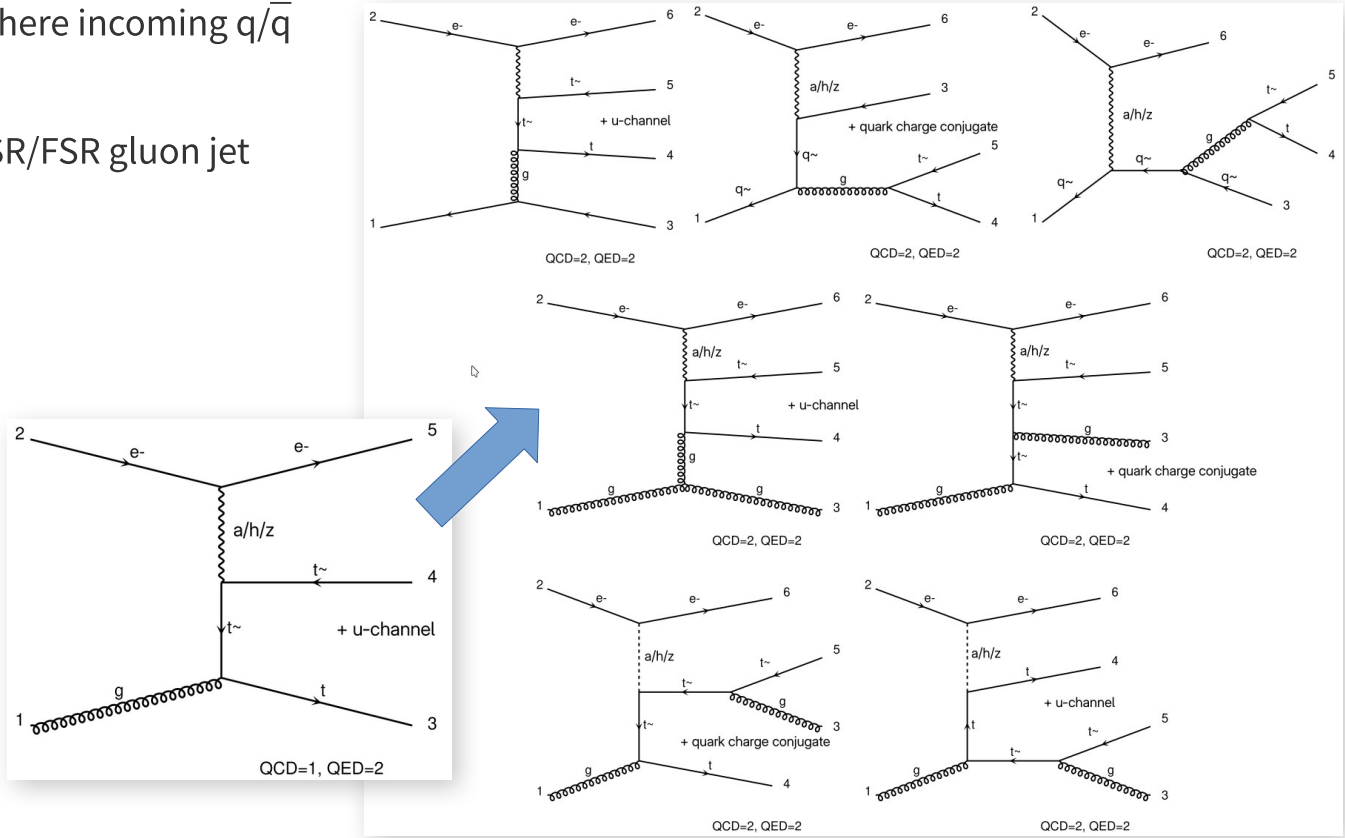
- Source of **original mismatch** between my (arXiv:2503.20475 [hep-ex]), Ali Can, and Hamzeh’s cross section values was identified and corrected:
  - As expected, including associated jet emission, large increase in the multiplicity of processes to the inclusive single-top and top pair production
  - e.g. for single-top production, additional ISR/FSR diagrams, but also gluon-induced production





Top pair production case

- For $t\bar{t}$ production, this effect is amplified
  - Adding in processes where incoming $q/\bar{q}$ are contributing
  - And obviously, extra ISR/FSR gluon jet emission diagrams



Numerical results (single-top & top pair)

- Using conditions consistent with the ones published in arXiv:2503.20475 [hep-ex]:
  - same MadGraph5_aMC@NLO version (v3.6.3), CTEQ1l PDF set, kinematic cuts "as inclusive as can be" ($p_T^{t/\bar{t}} > 0$, $|\eta^{t/\bar{t}}| < 6$)
- Large reduction in all (polarised + un-polarised) cross sections, effect much more (2 orders of magnitude) stringent for top pair production
- Effect of imposing a 0-jet veto over 1-jet diagrams is amplified for lower centre of mass energies though…

Proton energy, E_p		7 TeV			
Electron energy, E_e		50 GeV		20 GeV	
Electron polarisation, P_e		0	−0.8	0	−0.8
$\sigma(e^-p \to e^- t\bar{t} + \mathrm{X})$ [pb]	1-jet	2.1	2.2	0.45	0.48
	0-jet	0.024	0.025	0.0012	0.0012
	ratio	87.5	88	375	400
$\sigma(e^-p \to \nu_l \bar{t} + \mathrm{X})$ [pb]	1-jet	5.8	10.7	0.74	1.34
	0-jet	1.46	2.62	0.28	0.50
	ratio	4	4.1	2.6	2.7

1-jet / 0-jet

Numerical results (PDF sets comparison)

- Results in same ballpark as Hamzeh's (see https://indico.cern.ch/event/1625671/contributions/6852851) and Ali Can's (private communication)
  - Tensions due to the choice of PDF sets (CTEQ6l1, compatible with LHeC's CDR, or NNPDF23_nlo_as_0119, LHAPDF#230000)

Proton energy, E_p			7 TeV			
Electron energy, E_e			50 GeV		20 GeV	
Electron polarisation, P_e			0	−0.8	0	−0.8
$\sigma(e^- p \to e^- t\bar{t} + \mathrm{X})$ [pb]	Laurent	CTEQ6l1	0.024	0.025	0.0012	0.0012
	Ali Can		0.031	0.032	0.0014	0.0015
	ratio	NNPDF23	0.77	0.78	0.86	0.8
	Hamzeh			0.032		0.0015
	ratio			0.78		0.8
$\sigma(e^- p \to \nu_l \bar{t} + \mathrm{X})$ [pb]	Laurent	CTEQ6l1	1.46	2.62	0.28	0.50
	Ali Can		1.61	2.89	0.31	0.55
	ratio	NNPDF23	0.91	0.91	0.9	0.91
	Hamzeh			2.89		0.55
	ratio			0.91		0.91

Spares